

CHAPTER 5: DIVIDE-AND-CONQUER ALGORITHMS

[Binary Search: Finding a Book in an Alphabetized Book shelf](#)

[Quicksort: Splitting an Unsorted Pile of Books into Sorted Piles](#)

[Merge Sort: Merging Small Piles of Playing Cards into Larger Sorted Piles](#)

[Summing an Array of Integers](#)

[Karatsuba Multiplication](#)

[The Algebra Behind the Karatsuba Algorithm](#)

[Summary.](#)

[Further Reading](#)

[Practice Questions](#)

[Practice Projects](#)

CHAPTER 6: PERMUTATIONS AND COMBINATIONS

[The Terminology of Set Theory.](#)

[Finding All Permutations Without Repetition: A Wedding Seating Chart](#)

[Getting Permutations with Nested Loops: A Less-Than-Ideal Approach](#)

[Permutations with Repetition: A Password Cracker](#)

[Getting K-Combinations with Recursion](#)

[Get All Combinations of Balanced Parentheses](#)

[Power Set: Finding All Subsets of a Set](#)

[Summary.](#)

[Further Reading](#)

[Practice Questions](#)

[Practice Projects](#)